

MC111 - Mathematical, Algorithmic, and Computational Thinking

Semester: Autumn 2025-26

Instructor: Sunitha Vadivel Murugan

Office: Room 1212, Faculty Block-1

Office Hours: Tuesdays 15:30 – 16:30; I have an open-door policy during these scheduled hours, so please drop in without an appointment. If these times do not work for your schedule, please email me to arrange a separate meeting.

Email: v_suni@dau.ac.in

Class Schedule:

- **Official Structure:** 3 Lecture sessions + 1 Tutorial session per week.
 - **Course Format Note:** Please be aware that all four weekly sessions will follow the same interactive format. There is no practical difference between a "lecture" and a "tutorial" in this course; every class is a hands-on thinking lab.
 - **Sessions:** Tuesdays 11:00, Tuesdays 14:00, Thursdays 08:00, Fridays 11:00
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Course Description

This course is an introduction to the fundamental modes of thought that underpin mathematics and computer science. We will develop skills in mathematical, algorithmic, and computational thinking through an integrated curriculum of interactive puzzles, games, “maths magic” and hands-on activities. The goal is to build a powerful and versatile problem-solving toolkit by learning to think with precision, break down complex problems, identify patterns, and design logical solutions.

Course Methodology

This course is designed as a "thinking lab." As noted in the class schedule, all sessions are fully integrated and interactive. Every class will be a dynamic mix of introducing new concepts and immediately exploring them through puzzles, strategic games, and collaborative problem-solving. You should come to every class prepared to actively participate, experiment with ideas, and engage in discussion.

Learning Objectives

By the end of this course, students will be able to:

- **Analyse** complex problems by applying interwoven principles of abstraction, decomposition, and formal logic.
- **Design** and evaluate elegant and efficient step-by-step procedures (algorithms) for a variety of logical challenges.
- **Model** information and processes, representing them in structured ways to facilitate problem-solving.
- **Synthesize** mathematical, algorithmic, and computational strategies to develop holistic solutions, both individually and as part of a team.
- **Communicate** solutions and strategies clearly, collaborating effectively on group-based challenges.

Assessment and Grading Policy

Your final grade is a comprehensive measure of individual mastery and collaborative application. It is determined by regular assignments, two examinations, a final group project, and in-class participation.

- Weekly Problem Sets / Puzzle Solutions (15%)
- Hands-on Activity Write-ups (15%)
- Mid-Semester Examination (20%)

- End-Semester Examination (20%)
- Final Group Project (25%)
- Active Participation (5%)

Note on Grading Scheme: This course uses an absolute grading scheme. Your final grade will be awarded based on your cumulative score according to a pre-defined scale, not on a curve relative to the performance of other students in the class. The proposed grading scheme is

Grade Description	Outstanding	Excellent	Very Good	Good	Average	Below Average	Pass	Poor	Fail
Score Range (%)	91 - 100	81 - 90	71 - 80	61 - 70	51 - 60	46 - 50	40 - 45	35 - 39	< 35
Letter Grade	AA	AB	BB	BC	CC	CD	DD	DE	F
Grade Points	10	9	8	7	6	5	4	3	0

Note on Examinations: Both the Mid-Semester and End-Semester examinations will be conducted online through the Moodle LMS.

- **Mid-Semester Exam Syllabus:** Will cover all topics discussed in class from the beginning of the semester up to the last session before the exam.
- **End-Semester Exam Syllabus:** Will cover all topics discussed in class *after* the Mid-Semester examination.

Required Materials

There is no required textbook for this course. All necessary readings, papers, and puzzle descriptions will be presented during the class. You will be required to maintain a notebook for recording all of this and for your activity write-ups.

Recommended Technical Skills: While not mandatory, developing a basic understanding of HTML for formatting and LaTeX for typesetting mathematical notations will be a significant advantage for preparing your reports and write-ups. Excellent, free tutorials are available online:

- **For LaTeX:** The [Learn LaTeX in 30 minutes](#) interactive tutorial from Overleaf is a great place to start.
- **For HTML:** The [HTML Basics](#) guide from GeeksforGeeks provides a solid foundation.

Final Group Project: Design an Unplugged Activity

Project Goal: The goal of the final project is for student groups (3-4 members) to synthesize the three modes of thinking by designing, documenting, and demonstrating an original 'unplugged' activity, game, or puzzle that teaches a concept relevant to this course.

Project Components and Deliverables: The project grade is a significant component of your final mark and is comprised of the following parts, which include both group and individual submissions:

- **Public Exhibition (Group):** During the final week of the course, all groups will present their activity in a public exhibition. Your goal is to demonstrate the activity, engage visitors, and explain the core concepts it teaches.
- **Activity Kit (Group):** After the exhibition, each group will submit a complete, self-contained "activity kit" to SPARK. This kit must include all necessary physical materials and a clear instruction manual so that others can use the activity in the future.
- **Group Report (Group):** A written report analysing the core principles of the project. This report must detail the rules of the activity, the underlying mathematical, algorithmic, and computational concepts, and a discussion of your design process.
- **Individual Reflection (Individual):** Each student will submit an individual report exploring the principles of their group's project in greater depth. This may include proposing generalizations, analysing potential restrictions or edge cases, or connecting the core idea to other problems in mathematics and computer science.

Project Grading Breakdown: The final project component of your course grade will be calculated as follows:

- Public Exhibition & Demonstration: 30%
- Activity Kit & Instructions: 20%
- Group Report: 30%
- Individual Reflection: 20%

Project Proposal Guidelines: Your group must submit a 1-page project proposal for approval. The goal is to formalize your idea and receive feedback before you begin the main phase of your work. Your proposal must include the following sections:

1. **Project Title:** A creative and descriptive name for your activity, game, or puzzle.

- 2. Group Members:** A list of all members of your project group.
- 3. Core Concept:** A brief paragraph identifying the central concept from mathematics, computer science, or logic that your project will explore (e.g., graph colouring, recursive thinking, public-key cryptography, etc.).
- 4. Activity Overview:** A summary of the activity itself. What will a participant *do*? What is the goal of your game or the solution to your puzzle? How does it demonstrate the core concept?
- 5. Learning Objectives:** A list of 2-3 specific things a person should understand or be able to do after completing your activity.
- 6. Preliminary Materials:** A short, bulleted list of the physical materials you anticipate needing to create your "activity kit."
- 7. Any Additional Information:**

Key Dates & Deadlines

To help you plan your semester, here is a summary of key dates. These are based on academic calendar for Autumn 2025-26, and may be subject to minor changes.

Date(s)	Milestone
Oct 3 - 10	Mid-Semester Examination (exact date TBD)
Oct 20	Project Proposal Due (1-page summary)
Nov 15-20	Public Project Exhibition (Date TBD) All Project Materials Due (Date TBD)
Dec 1 - 6	End-Semester Examination (exact date TBD)

Code of Conduct & Academic Integrity

All students are expected to adhere to the highest standards of academic honesty and classroom conduct, as outlined in the university's official student handbook.

Classroom Conduct: A productive learning environment is built on mutual respect. You are expected to actively participate in activities, engage in discussions thoughtfully, and listen to the perspectives of your peers and instructor.

Academic Integrity & Plagiarism: While collaboration on in-class activities and puzzles is highly encouraged, all submitted work—including problem sets, write-ups, reports, and examinations—must be the result of your own individual effort and understanding.

Plagiarism is defined as the act of using another person's ideas, words, or work and presenting them as your own without proper citation or acknowledgement. This includes copying from online sources, other students, or previous course materials. Any instance of academic dishonesty will be taken very seriously and will result, at a minimum, in a score of zero for the assignment and may be escalated to the university's disciplinary committee. When in doubt about how to properly cite or collaborate, please ask the instructor.

Policy on the Use of Artificial Intelligence (AI) Tools

Our Guiding Philosophy: The primary goal of this course is to develop *your* unique mathematical, algorithmic, and computational thinking skills. The assignments are designed to be the practice ground for this development. Therefore, our policy on AI tools like ChatGPT, Gemini, Copilot, etc., is designed to protect this core objective.

Prohibited Uses: The following uses of AI tools are strictly forbidden and will be treated as a violation of academic integrity:

- Generating solutions for any graded problem set, puzzle, or activity write-up.
- Writing or generating any portion of your individual or group reports.
- Using AI tools in any capacity during the mid-semester or end-semester examinations.

Submitting work generated by an AI as your own is a form of plagiarism and will be subject to the consequences outlined in the Code of Conduct.

Permitted & Encouraged Uses: You are permitted and even encouraged to use AI tools in the following ways to support your learning:

- **As a brainstorming partner:** To explore related topics or ask for alternative explanations of a concept we've discussed in class (e.g., "Can you explain the concept of recursion using an analogy other than the Tower of Hanoi?").
- **As an editing assistant:** To check for grammar, spelling, and clarity on a report *you have already written*.
- **As a study tool:** To generate practice questions or quiz yourself on key terms and definitions.

Requirement for Disclosure: If you use an AI tool for any of the permitted purposes on an assignment, you must include a brief paragraph at the end of your submission detailing which tool you used and for what purpose. Transparency is key to responsible use.

A Final Word of Caution: You are responsible for the correctness and integrity of everything you submit. AI tools can and do make mistakes. Relying on them for understanding rather than for support will hinder your learning and likely impact your performance.

[Assignment Submission Policy](#)

All assignments, including problem sets and activity write-ups, must be submitted electronically through the Moodle LMS. To ensure fairness and keep all students on the same schedule, late submissions for problem sets and activity write-ups will not be accepted.

To provide flexibility and account for illness, emergencies, or other life events, I will automatically drop your lowest scores at the end of the semester. For example, if we have 10 problem sets, only your best 8 scores will count toward your final grade (i.e., your lowest two scores will be dropped). The same principle will apply to the activity write-ups. Because this flexible policy is in place, no individual extensions will be granted. It is your responsibility to manage your submissions accordingly.

[Instructor Communication Policy](#)

The best way to contact me is via email. I will do my best to respond within 48 hours on weekdays. Emails sent over the weekend will typically be answered on the following Monday. Ensure that you sign-off your emails with your name.

Tentative Topical Outline (Integrated Approach)

Our journey is structured around core thinking skills. Each week, we will use interactive puzzles, games, and challenges as a vehicle to explore these powerful concepts.

- **Weeks 1-2: The Art of Precise Instruction & Logical Deduction**

Concepts: We will explore the foundations of rigorous thinking, investigating logical paradoxes, the importance of precise instructions, and how to deduce truths from constrained information.

- **Weeks 3-4: Thinking with Graphs and Networks**

Concepts: This module focuses on modelling relationships. We will explore how to represent networks, find efficient paths, and understand the principles of connectivity and structure through hands-on challenges.

- **Weeks 5-6: Order, Searching, and Algorithm Efficiency**

Concepts: How do we create order out of chaos? We will physically simulate and analyse various strategies for sorting information, searching for items, and measuring the efficiency of our methods.

- **Week 7: Mid-Semester Examination**

- **Weeks 8-9: Information, Representation, and Secret Sharing**

Concepts: This module explores how information is represented and secured. We will delve into the mathematics of secret-keeping through classic cryptography and learn how data can be encoded for different purposes.

- **Weeks 10-11: Recursive Thinking and Strategic Gaming**

Concepts: We will tackle problems that contain smaller versions of themselves, exploring the elegant and powerful idea of recursion. We will also investigate the logic of game theory and how to identify winning strategies.

- **Weeks 12-13: Handling Complexity, Concurrency, and Infinity**

Concepts: Here we confront the limits of computation. We will grapple with famously "hard" problems, the challenges of simultaneous processes, and the mind-bending nature of infinite sets through famous thought experiments.

- **Week 14: Advanced Topics in Proof and Computation**

Concepts: Our final module explores the frontiers of logic, including how it's possible to prove you know a secret without revealing the secret itself, and we will dedicate time to developing your final projects.

- **Finals Week: Final Group Project Due & End-Semester Examination**

A Note on Discovery and Spoilers: The magic of this course comes from live, collaborative problem-solving. The puzzles and activities are designed to be tackled with fresh eyes. To preserve this experience for yourself and your classmates, you are strongly encouraged **not** to search for the solutions to any in-class activities or puzzles ahead of time or until asked to do so. The goal is not just to find the answer, but to experience the journey of discovery.

Food for Thought: The Spirit of Our Class

“A course this unique is hard to capture in a single phrase,” say the students. As a gift to the new comers, the seniors brainstormed many possible mottos and taglines to capture the various flavours of this course as a way to describe the course. Here is a collection (listed alphabetically) of those ideas. By the end of the semester, you might even have a few of your own!

The grand list:

- A Boot Camp for Your Brain.
- A Course in Mathematical and Computational Thinking.
- A Course in Pure Reason.
- A Course That Leaves a Legacy.
- A Hands-on Approach to Foundational Concepts.
- A Practical Guide to Abstract Thinking.
- A User Manual for Your Mind.
- All Killer, No Filler, No Computer.
- All the fun of a game night, with the academic rigor of a final exam.
- An Archaeology of Algorithms.
- An Introduction to Computing Without Computers
- An Introduction to Lamport's Bakery Algorithm... Using People.
- Analog Logic in a Digital World.
- Because 'brute force' is not an acceptable life strategy.
- Before ChatGPT: Master the Original Intelligence.
- Before the "Hello, World!", There Was the Thought.
- Before There Were Computers, There Was Computation.
- Beyond P and NP: A Practical Course in Hard Problems.
- Beyond the Blockchain: The Logic of Distributed Trust.
- Beyond the Keyboard: Thinking Made Tangible.
- Brains & Machines: A Thinking Course for the Digital Age
- Build a Human Turing Machine. Touch the Edge of Computability.
- Building a better way to solve problems.
- Building a Problem-Solver's Mind.
- Building the Problem-Solver's Mindset.
- By the end of this, you'll be able to explain recursion to your parents. (Results not guaranteed).
- Can a Barber Shave Himself? Find Out Inside.
- Card Tricks, Cryptography, and the Secrets of Computation.
- Code is for Computers. Logic is for People.
- Come for the algorithms, stay for the arts and crafts.
- Come for the Card Tricks, Stay for the Computability Theory.

- Come for the card tricks, stay for the existential dread about computability.
- Come for the easy credits. Just kidding. Run.
- Come for the fun activities. Stay for the character-building exams.
- Come for the logic, stay because you're hopelessly stuck on the Knight's Tour.
- Come for the Puzzles. Leave with a Superpower.
- Come in with problems. Leave with more, but much more interesting, problems.
- Come write our final exam. We promise it's more fun than a bug-riddled compiler.
- Computation is Applied Mathematics. Let's Prove It.
- Computing Without Computers: It's Not Magic, It's Logic.
- Computing without Computers: Think Before You Code.
- Congratulations, you've solved all the puzzles! The final boss is the exam.
- Deconstruct Problems. Rebuild Your Mind.
- Don't Just Scroll the Feed. Understand the Algorithm That Builds It.
- Don't Just Solve the Puzzle. Build It.
- Don't Just Use the System. Understand It.
- Engineering from First Principles.
- Ever Wanted to Win at Nim? Now You Can.
- Exploring the Core of Computation, Unplugged.
- F is not a grade, it's a lifestyle.
- Final Project: One part logic, one part craft supplies.
- Finally, a class where you can't just Google the answer. We checked.
- Forge Your Logic. Frame Your Future.
- Forget "Hello, World!"—Our First Program is the Tower of Hanoi.
- Formalizing Intuition.
- From Abstract Logic to Tangible Artifact.
- From 'Aha!' to Algorithm.
- From Ancient Puzzles to Modern Machines
- From Concrete Puzzles to Abstract Principles.
- From Puzzles to Proofs, from Ideas to Algorithms.
- From Puzzles to Proofs: A New Way to Think.
- From Sorting Cards to Analyzing Complexity.
- From the Knight's Tour to Graph Theory.
- From the Tower of Hanoi to understanding recursion.
- Future-Proof Your Mind.
- Go Beyond Coding: A Course in Design, Demonstration, and Pedagogy.
- Guaranteed to make you the most insufferably logical person at any party.
- Hacking Your Own Brain: No Keyboard Required.

- Hardware is for Kicking. Software is for Thinking.
- Hats, Hanoi, and the Halting Problem.
- Hilbert's Hotel is Full. We Have a Room for You.
- How Ideas Become Algorithms, and Algorithms Become Code
- How to Solve It — with Math, Code, and Logic
- I was told there would be extra credit.
- Intelligence, Unplugged.
- It's all fun and games until someone mentions the Halting Problem.
- It's Not a Card Trick, It's an Algorithm.
- It's not magic, it's logic. (But we'll teach you the magic tricks anyway).
- Languages Fade. Logic is Forever.
- Learn It. Build It. Teach It.
- Learn One Language to Rule Them All: The Language of Problems.
- Learn to Think Like a Computer Scientist, Not Just Use a Computer.
- Learn to think, not just to code.
- Less lecture, more logic puzzles.
- Logic and Math.
- Logic Puzzles, Zero-Knowledge Proofs, and Everything In Between.
- Logic, Distilled.
- Master the How. The What Will Follow.
- Mastering Computation, No Power Cord Required.
- Mastermind, Minimal Spanning Trees, and the Mathematics of Thought.
- MC111: The Course.
- More Math than the CS Majors, More CS than the Math Majors.
- More than just card tricks: A course in algorithms.
- More write-ups than sleep.
- My brain has too many tabs open.
- My final project is this tagline.
- Our exams are just like the class, but with more crying.
- Our final exam is just one big escape room. The room is your own mind.
- Our only prerequisite is a brain. (Batteries not included).
- Play Games. Solve Puzzles. Accidentally Master Computer Science.
- Play the Nim Game. Discover Winning Strategies.
- Power Up Your Brain, Not Your Laptop.
- Prove You Understand It. By Proving You Can Explain It.
- Puzzles, Games, and the Principles of Computation
- Side effects may include winning every argument and losing every friend.
- Solve the Barber's Paradox. Master Formal Logic.
- Solving puzzles, learning logic.
- Speak Fluent Logic.

- Sprouts, Spots, and Spanning Trees: An Introduction to Algorithmic Fun.
- Strip Away the Syntax. Master the Science.
- The 'And' in Mathematics and Computing.
- The Architecture of a Correct Thought.
- The Art and Science of Problem Solving.
- The Art of Thinking: Algorithms, Logic, and You.
- The best way to learn is to teach. Welcome to your final exam.
- The class that makes your other classes make sense.
- The Class That Teaches You to Build the Demo.
- The Class Your Future Boss Wishes They Had Taken.
- The Elegant Machinery of Thought.
- The Final Exam is an Exhibition, Not an Examination.
- The Foundations of Innovation
- The Grammar of a Good Solution.
- The hard fun.
- The Hardest Fun You'll Have All Semester.
- The Hidden Code Behind Smart Thinking
- The Human Side of Computation.
- The Laws of Thought.
- The Logic Behind the Language Model.
- The Metalanguage of Machines.
- The mid-sem exam: Now with 50% more mind-bending puzzles and 100% less mercy.
- The most fun you can have while your brain is overheating.
- The Most Important Course You'll Ever Take, Disguised as the Most Fun.
- The Most Powerful Tool is the One Between Your Ears. Let's Sharpen It.
- The One Class That Makes Every Other Class Easier.
- The One Skill That Will Never Be Deprecated.
- The Only B.Tech Course With a Hot Glue Gun on the Syllabus.
- The only class where "playing games" is the lecture.
- The only course that will leave you questioning the logical consistency of a sandwich.
- The only exam you'll ever take where the first question is a paradox.
- The only final exam where you invent the questions.
- The Physics of Computation.
- The reason I now overthink everything.
- The reason your other classes will suddenly feel easy.
- The Source Code of a Good Idea.
- The Thinker's Toolkit for a Coded World.
- The Trojan Horse of Your Curriculum.

- The Ultimate Mental OS: Unplugged Thinking for a Digital Age.
- The Ultimate Prompt Engineering: How to Ask the Right Questions.
- The Universal Assembly Language... For Your Brain.
- Think Different. Think Deeper.
- Think in Code, Reason in Logic, Win with Puzzles.
- Think Like a Computer (Scientist)
- Think Like a Hacker (The Original Kind).
- Think With Clarity. Build With Purpose.
- Thinking Clearly in the Age of Computation
- Thinking Skills for the Digital Age.
- Thinking without a computer.
- This class has no bugs, only undocumented features of your brain.
- This Course Will Rewire Your Thinking. Guaranteed.
- This Semester, You're Not Just a Student. You're a Game Designer.
- Today's agenda: The Nim Game, the Tower of Hanoi, and breaking your brain.
- Train your brain to think like a mathematician, design like an algorithmist, and reason like a computer — all in one course.
- Train Your Own Neural Network: Your Brain.
- Understanding the Logic Behind the Code.
- Unplugged: The Algorithms are in Your Head, Not the Machine.
- Warning: Final project may involve more cardboard and glitter than code.
- Warning: May Cause Spontaneous Problem-Solving.
- Warning: May lead to spontaneously organizing your room into a hash table.
- Warning: May lead to spontaneously organizing your sock drawer into a hash table.
- We play with cards, solve paradoxes, and build human computers. Then we test you on it.
- We run on $O(\log n)$... where 'n' is the number of brain cells you have left.
- What Color is Your Hat? The Answer is Logic.
- Where formal Zero-Knowledge Proofs are Demonstrated, Not Just Described.
- Where Mathematics Ends and Computation Begins.
- Where sorting cards and coloring maps is how we study complexity.
- Where Your Grade Depends on How Well You Can Explain a Game to a Confused Visitor.
- Where Zero-Knowledge Proofs are Demonstrated, Not Just Described.
- While They Learn to Code, You Will Learn to Think.
- Yes, the activities are fun. No, that won't be on the exam. Or will it?

- You Can't Google This Kind of Understanding.
- You'll be too busy having fun to realize you're learning.
- You'll love the puzzles. You'll love the games. You'll have... strong opinions... about the exam.
- Your Brain's API: Accessing a Better Way to Think.
- Your Final Project Will Be Our Next Homework.
- Your OS Will Change. Your Thinking Won't Have To.
- Your Unfair Advantage.